

220B Homework 3

Due: Sunday, 24 May

Problem 1: Risk-set geometry and a randomized minimax rule (25 pts)

Consider a two-person zero-sum game with $\Theta = \{\theta_1, \theta_2\}$, $\mathcal{A} = \{a_1, a_2, a_3\}$, and loss matrix

	a_1	a_2	a_3
θ_1	3	1	0
θ_2	1	2	4

- (a) (5) Plot the three nonrandomized risk points and describe the risk set $S \subseteq \mathbb{R}^2$ explicitly. Identify the lower boundary $\lambda(S)$.
- (b) (5) Determine which nonrandomized actions are admissible.
- (c) (8) Find the minimax risk point $s_M \in S$ by locating where the lower quadrant $Q_\alpha = \{z : z_i \leq \alpha\}$ first meets S as α decreases. State the value V , express s_M as a convex combination of nonrandomized risk points, and state the corresponding minimax randomized strategy δ_M^* for Player II.
- (d) (7) Identify a maximin strategy π_M for Player I from the tangent-hyperplane characterization. Verify directly that $L(\theta, \delta_M^*) \leq V \leq L(\pi_M, a)$ for every $\theta \in \Theta$ and every $a \in \mathcal{A}$.

Hint: Let E^* be the edge of $\lambda(S)$ containing s_M . The inward normal of E^* has both components nonnegative; normalizing it to a probability vector yields the maximin prior.

Problem 2: Minimax for the binomial proportion (30 pts)

Let $X \sim \text{Binomial}(n, \theta)$ with $\theta \in [0, 1]$, action space $\mathcal{A} = [0, 1]$, and squared-error loss $L(\theta, a) = (\theta - a)^2$.

(a) (6) For the prior $\pi = \text{Beta}(\alpha, \beta)$ with $\alpha, \beta > 0$, derive the Bayes rule $\delta_{\alpha, \beta}(X)$ under squared-error loss.

(b) (8) Compute $R(\theta, \delta_{\alpha, \beta})$ as an explicit polynomial in θ over the denominator $(\alpha + \beta + n)^2$.

(c) (10) Find the unique values (α^*, β^*) with $\alpha^*, \beta^* > 0$ for which $R(\theta, \delta_{\alpha^*, \beta^*})$ does not depend on θ . Compute the constant value V in closed form, and state the resulting equalizer rule $\delta^* := \delta_{\alpha^*, \beta^*}$.

(d) (6) Conclude that δ^* is minimax over the class of all (possibly randomized) decision rules. Identify the maximin prior.

Hint: For (c), require the coefficients of θ^2 and θ in the numerator from (b) to vanish, giving two equations in α and β . For (d), sandwich $\sup_{\theta} R(\theta, \delta^*)$ between two Bayes-risk quantities using the constant-risk property and the Bayes-optimality of δ^* against the maximin prior.

Problem 3: Minimax via a sequence of priors (20 pts)

Let $X \sim N(\theta, 1)$ with $\theta \in \mathbb{R}$, and squared-error loss $L(\theta, a) = (\theta - a)^2$. Let $\delta_0(X) = X$.

(a) (3) Compute $R(\theta, \delta_0)$ and $\sup_{\theta \in \mathbb{R}} R(\theta, \delta_0)$.

(b) (8) For each $k > 0$, let $\pi_k = N(0, k)$. Derive the Bayes rule $\delta_{\pi_k}(X)$ under squared-error loss, and compute the minimum Bayes risk $\inf_{\delta} r(\pi_k, \delta) = r(\pi_k, \delta_{\pi_k})$ in closed form.

(c) (9) Apply Theorem 3 of Lecture 6 to the sequence $\{\pi_k\}_{k \geq 1}$ to conclude that δ_0 is minimax with value $V = 1$. Explain why no single proper prior π on $\mathbb{R}$ suffices for this conclusion, but a sequence of proper priors does.

Hint: For (c), the verification reduces to showing $\sup_{\theta} R(\theta, \delta_0) \leq \liminf_k r(\pi_k, \delta_{\pi_k})$. The Bayes risk $r(\pi_k, \delta_{\pi_k})$ equals the (constant) posterior variance from (b).

Problem 4: Restricted-parameter minimax via a two-point prior (25 pts)

Let $X \sim N(\theta, 1)$ with $\theta \in [-m, m]$ for some fixed $m > 0$, and squared-error loss $L(\theta, a) = (\theta - a)^2$.

(a) (8) Consider the two-point prior $\pi_m = \frac{1}{2}\delta_{-m} + \frac{1}{2}\delta_m$. Derive the Bayes rule $\delta_{\pi_m}(X)$ under squared-error loss in closed form, and show that $\delta_{\pi_m}(X) = m \tanh(mX)$.

(b) (6) Show that $R(m, \delta_{\pi_m}) = R(-m, \delta_{\pi_m})$ by a symmetry argument.

(c) (6) Suppose that $\sup_{\theta \in [-m, m]} R(\theta, \delta_{\pi_m}) = R(m, \delta_{\pi_m})$ (this holds for all m sufficiently small; you may assume it without proof). Verify that δ_{π_m} is minimax over the restricted parameter space $\Theta = [-m, m]$. State the minimax value.

(d) (5) As $m \rightarrow \infty$, what does $\delta_{\pi_m}(X) = m \tanh(mX)$ converge to (pointwise in X)? Compare with the minimax rule on the unrestricted parameter space $\Theta = \mathbb{R}$ from Problem 3, and comment on why one cannot recover the unrestricted minimax rule by sending $m \rightarrow \infty$.

Hint: For (a), multiply numerator and denominator of the posterior probabilities by $\exp((X^2 + m^2)/2)$ to reduce to e^{mX}, e^{-mX} . For (b), substitute $X = -Y$ in the integral defining $R(-m, \delta_{\pi_m})$ and use that $\tanh$ is odd.